1 Linear systems

REF zero rows at the bottom; leading entries move strictly right. **RREF** also: each leading entry is 1 and alone in its column. RREF is unique, REF is not.

Row ops $R_i \leftrightarrow R_j$, cR_i $(c \neq 0)$, $R_i + cR_j$. $2R_2 + R_1$ **is two ops**, not one.

Solve $Ax = b$ ❶ $(A \mid b) \to$ REF. ❷ Pivot in the last column $\Rightarrow$ **inconsistent**. ❸ Otherwise every non-pivot column of A gets a parameter $s, t, \ldots$; back-substitute. Unique $\Leftrightarrow$ no non-pivot column (forces $m \geq n$). Number of parameters $= n - r$.

Solution structure general solution = one particular solution + general solution of $Ax = 0$. In $x = p + sv_1 + tv_2$: $Ap = b$, $Av_i = 0$, nullity 2. Homogeneous: always consistent; $n > m \Rightarrow$ nontrivial solutions.

Unknown constants never divide or scale by an expression that may be 0. Reduce, then split into cases where a would-be pivot vanishes. Square A: $\det \neq 0 \Leftrightarrow$ unique; then substitute each root of $\det = 0$ and reduce again.

System from its solution set $p + sv_1 + tv_2$: coefficient rows a solve $a \cdot v_i = 0$ for all i; right-hand sides are $a \cdot p$.

2 Matrices, inverses, LU

Algebra $(AB)^{\mathrm{T}} = B^{\mathrm{T}}A^{\mathrm{T}}$, $(AB)^{-1} = B^{-1}A^{-1}$, $(A^{\mathrm{T}})^{-1} = (A^{-1})^{\mathrm{T}}$. $Ax = \sum x_j a_j$; column j of AB is Ab_j. Upper × upper is upper; symmetric and triangular $\Rightarrow$ diagonal.

False in general: $AB = BA$; $AB = 0 \Rightarrow A = 0$ or $B = 0$; $AB = AC \Rightarrow B = C$ (needs A invertible); $(A + B)^2 = A^2 + 2AB + B^2$; A, B invertible $\Rightarrow A + B$ invertible; symmetric × symmetric is symmetric; non-square $AB = I \Rightarrow BA = I$.

Inverse $(A \mid I) \to (I \mid A^{-1})$; a zero row on the left $\Rightarrow$ singular. Solve $AX = B$ by $(A \mid B) \to (I \mid X)$. $A^{-1} = \operatorname{adj}(A)/\det A$, where $\operatorname{adj}(A)_{ij} = A_{ji}$.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Elementary matrix E is I after **one** row op; EA does that op to A.

Row op	Undo	$\det E$ = factor on $\det A$
$R_i \leftrightarrow R_j$	$R_i \leftrightarrow R_j$	-1
cR_i	$(1/c)R_i$	c
$R_i + cR_j$	$R_i - cR_j$	1

$E_k \cdots E_1 A = I \Rightarrow A^{-1} = E_k \cdots E_1$ and $A = E_1^{-1} \cdots E_k^{-1}$.

LU ❶ $A \to U$ (REF) using only $R_i + cR_j$ with $i > j$: no swaps, no scaling. ❷ L: 1s on the diagonal and $-c$ in entry (i, j), so $R_3 - 2R_2$ puts $+2$ at $(3, 2)$. ❸ $Ax = b$: solve $Ly = b$ downwards, then $Ux = y$ upwards. A forced swap $\Rightarrow$ no LU (but $PA = LU$ for a permutation P).

Given $A = LU$ A is row equivalent to U: same rank, pivot positions, Null, Row, and the same relations between columns ($u_2 = 2u_1 \Leftrightarrow a_2 = 2a_1$). Basis of $\operatorname{Col}(A)$: $L \times$ (pivot columns of U), **not** the columns of U. Square: $\det A = \prod u_{ii}$.

2 Determinants

Cofactor expansion along any row or column: $\det A = \sum_j a_{ij}A_{ij}$, $A_{ij} = (-1)^{i+j} \det M_{ij}$. 2×2: $ad - bc$. 3×3: $aei + bfg + cdh - ceg - afh - bdi$. Triangular: product of the diagonal.

By reduction to a triangular R, using the table above: $\det A = \pm(\prod r_{ii})/(\prod$ of the scalings $c)$, with $-$ for an odd number of swaps. Column ops act the same way.

Properties $\det(AB) = \det A \det B$ ▪ $\det A^{\mathrm{T}} = \det A$ ▪ $\det A^{-1} = 1/\det A$ ▪ $\det(cA) = c^n \det A$ ▪ $\det(\operatorname{adj} A) = (\det A)^{n-1}$ ▪ $A \operatorname{adj}(A) = (\det A)I$ ▪ proportional rows or a zero row $\Rightarrow 0$ ▪ $\det(A + B) \neq \det A + \det B$.

Cramer $x_i = \det(A_i)/\det A$; A_i is A with column i replaced by b. **Area** of the parallelogram on $u, v \in \mathbb{R}^2$: $|\det(u\ v)|$; triangle: half.

Pattern determinants $(n \times n)$ one row op to reach a triangular matrix, or expand along row 1 for a recurrence in d_n. EG a on the diagonal, b in the two far corners: $R_n - (b/a)R_1$ gives $\det = a^{n-2}(a^2 - b^2)$.

3 Span, independence, basis

In the span? $v \in \operatorname{span}\{u_1, \ldots, u_k\} \Leftrightarrow (u_1 \cdots u_k \mid v)$ is consistent; the solution gives the coefficients. Several vectors at once: $(u_1 \cdots u_k \mid v_1 v_2 \cdots)$.

Spans $\mathbb{R}^n \Leftrightarrow$ REF of $(u_1 \cdots u_k)$ has no zero row (rank n); impossible when $k < n$.

Span inclusion $\operatorname{span} T \subseteq \operatorname{span} S \Leftrightarrow (S \mid T)$ is consistent in every column; equal $\Leftrightarrow \subseteq$ and same dimension.

Subspace? **Yes:** write it as a span, or as $\operatorname{Null}(A)$ (homogeneous linear equations). **No:** $0 \notin V$, or show $u, v \in V$ with $u + v \notin V$ (or $cu \notin V$). Red flags: a nonzero constant, products or squares ($xy = 0$, $x^2 = y^2$), "or", inequalities. $U \cap V$ and $U + V$ are subspaces, $U \cup V$ **usually is not**. $\dim(U + V) = \dim U + \dim V - \dim(U \cap V)$.

Independent? $\Leftrightarrow (u_1 \cdots u_k)x = 0$ only trivially $\Leftrightarrow$ every column is a pivot column (rank k). Dependent at sight: $k > n$; contains 0; two parallel vectors. Independence survives: taking subsets; adding a vector outside the span; passing to coordinates $[u_i]_S$ or to Pu_i with P invertible.

Basis of V = independent and spans V. If $\dim V = k$: any k vectors of V that are independent (or span V) are a basis. dim: $\mathbb{R}^n$ is n; $\{a \cdot x = 0\}$ is $n - 1$; $\operatorname{Null}(A)$ is $n - r$; span S is the rank of $(u_1 \cdots u_k)$. $U \subseteq V \Rightarrow \dim U \leq \dim V$, equal $\Leftrightarrow U = V$. $\{0\}$ has basis $\varnothing$, dim 0; $\varnothing$ is independent, $\{0\}$ is dependent.

New basis from old $\{u_i\}$ a basis, $v_j = \sum_i c_{ij}u_i$: $\{v_j\}$ is a basis $\Leftrightarrow C = (c_{ij})$ is invertible.

Coordinates $[v]_S$: solve $(u_1 \cdots u_k \mid v)$. Linear: $[au + bv]_S = a[u]_S + b[v]_S$.

Transition matrix from $T = \{v_j\}$ to S $(S \mid T) \to (I \mid P)$, so $P = ([v_1]_S \cdots [v_k]_S)$. Then $[w]_S = P[w]_T$, and from S to T it is P^{-1}.

4 Row, column, null space · rank

Space	in	Basis, after $A \to R$	dim
$\operatorname{Row}(A)$	$\mathbb{R}^n$	nonzero rows of R	r
$\operatorname{Col}(A)$	$\mathbb{R}^m$	columns of A at the pivots	r
$\operatorname{Null}(A)$	$\mathbb{R}^n$	$Ax = 0$: a vector per parameter	$n - r$
$\operatorname{Null}(A^{\mathrm{T}})$	$\mathbb{R}^m$	solve $A^{\mathrm{T}}y = 0$; it is $\operatorname{Col}(A)^{\perp}$	$m - r$

Row ops keep Row, Null, rank and the relations between columns, **not** Col. In the RREF a non-pivot column lists its coefficients on the pivot columns.

Basis chosen from a set vectors as columns, keep those in pivot positions. Rows of A forming a basis of $\operatorname{Row}(A)$: do it to A^{T}. **Extend to a basis of** $\mathbb{R}^n$ vectors as rows $\to$ REF $\to$ add e_j for each non-pivot column j.

Same row space (equal sizes) $\Leftrightarrow$ same RREF. **Consistent** $Ax = b \Leftrightarrow b \in \operatorname{Col}(A) \Leftrightarrow \operatorname{rank} A = \operatorname{rank}(A \mid b)$.

Rank facts $\operatorname{rank} A = \operatorname{rank} A^{\mathrm{T}} \leq \min(m, n)$ ▪ **rank + nullity** $= n$ ▪ $\operatorname{rank}(AB) \leq \min(\operatorname{rank} A, \operatorname{rank} B)$ ▪ $\operatorname{rank}(A + B) \leq \operatorname{rank} A + \operatorname{rank} B$ ▪ $\operatorname{rank}(A^{\mathrm{T}}A) = \operatorname{rank} A$ and $\operatorname{Null}(A^{\mathrm{T}}A) = \operatorname{Null}(A)$ ▪ $\operatorname{Col}(AB) \subseteq \operatorname{Col}(A)$, $\operatorname{Null}(B) \subseteq \operatorname{Null}(AB)$ ▪ $AB = 0 \Leftrightarrow \operatorname{Col}(B) \subseteq \operatorname{Null}(A)$ ▪ P invertible: $\operatorname{rank}(PA) = \operatorname{rank}(AP) = \operatorname{rank} A$.

Rank with parameters reduce without dividing by parameter expressions; rank = number of nonzero rows, so split into cases where a would-be pivot is 0. Nullity $= n -$ rank.

Full rank $r = n$ (columns indep.)	**Full rank** $r = m$ (rows indep.)
$Ax = b$: at most one solution	$Ax = b$: consistent for every b
$\operatorname{Null}(A) = \{0\}$; needs $n \leq m$	$\operatorname{Col}(A) = \mathbb{R}^m$; needs $m \leq n$
$A^{\mathrm{T}}A$ invertible	AA^{T} invertible
left inv. $(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}$	right inv. $A^{\mathrm{T}}(AA^{\mathrm{T}})^{-1}$
$x \mapsto Ax$ injective	$x \mapsto Ax$ surjective

All right inverses solve $AX = I$: reduce $(A \mid I)$ and give the general solution column by column. **All left inverses** are the right inverses of A^{T}, transposed. Square: both are A^{-1}.

Square A of order n is invertible $\Leftrightarrow$ RREF is $I \Leftrightarrow$ product of elementary matrices $\Leftrightarrow \det A \neq 0 \Leftrightarrow Ax = 0$ only trivially $\Leftrightarrow Ax = b$ has a unique solution for every $b \Leftrightarrow$ a left or a right inverse exists $\Leftrightarrow A^{\mathrm{T}}$ invertible $\Leftrightarrow$ columns (rows) independent $\Leftrightarrow$ they span $\mathbb{R}^n \Leftrightarrow$ rank $n \Leftrightarrow$ nullity 0 $\Leftrightarrow 0$ is not an eigenvalue $\Leftrightarrow x \mapsto Ax$ injective $\Leftrightarrow$ surjective.

Notation A is $m \times n$ with columns a_j · $r = \operatorname{rank} A$ · R is a row-echelon form of A · vectors are columns · chips give the lecture chapter

5 Orthogonality and projection

Basics $\boldsymbol{u}\cdot\boldsymbol{v}=\boldsymbol{u}^{\mathrm{T}}\boldsymbol{v}$, $\|\boldsymbol{u}\|=\sqrt{\boldsymbol{u}\cdot\boldsymbol{u}}$, $d(\boldsymbol{u},\boldsymbol{v})=\|\boldsymbol{u}-\boldsymbol{v}\|$, $\cos\theta=\boldsymbol{u}\cdot\boldsymbol{v}/(\|\boldsymbol{u}\|\|\boldsymbol{v}\|)$. $\|\boldsymbol{u}+\boldsymbol{v}\|^2=\|\boldsymbol{u}\|^2+2\boldsymbol{u}\cdot\boldsymbol{v}+\|\boldsymbol{v}\|^2$, so Pythagoras $\Leftrightarrow\boldsymbol{u}\cdot\boldsymbol{v}=0$; $\|\boldsymbol{u}+\boldsymbol{v}\|\le\|\boldsymbol{u}\|+\|\boldsymbol{v}\|$. Orthogonal **nonzero** vectors are independent.

Orthogonal complement put vectors spanning V as the columns of $\boldsymbol{A}$: $V^{\perp}=\mathrm{Null}(\boldsymbol{A}^{\mathrm{T}})$. $\dim V+\dim V^{\perp}=n$, $(V^{\perp})^{\perp}=V$, $V\cap V^{\perp}=\{\boldsymbol{0}\}$. $\mathrm{Row}(\boldsymbol{A})^{\perp}=\mathrm{Null}(\boldsymbol{A})$, $\mathrm{Col}(\boldsymbol{A})^{\perp}=\mathrm{Null}(\boldsymbol{A}^{\mathrm{T}})$.

Orthogonal basis $\{\boldsymbol{u}_i\}$ coordinates with no row reduction: $c_i=(\boldsymbol{v}\cdot\boldsymbol{u}_i)/(\boldsymbol{u}_i\cdot\boldsymbol{u}_i)$. Orthonormal S: $[\boldsymbol{x}]_S\cdot[\boldsymbol{y}]_S=\boldsymbol{x}\cdot\boldsymbol{y}$ and $\|[\boldsymbol{x}]_S\|=\|\boldsymbol{x}\|$ **(false if merely orthogonal)**.

Projection $\boldsymbol{w}_p$ of $\boldsymbol{w}$ onto V

- **orthogonal** basis $\{\boldsymbol{u}_1,\dots,\boldsymbol{u}_k\}$:
$$\boldsymbol{w}_p=\frac{\boldsymbol{w}\cdot\boldsymbol{u}_1}{\boldsymbol{u}_1\cdot\boldsymbol{u}_1}\boldsymbol{u}_1+\cdots+\frac{\boldsymbol{w}\cdot\boldsymbol{u}_k}{\boldsymbol{u}_k\cdot\boldsymbol{u}_k}\boldsymbol{u}_k$$
- any basis as columns of $\boldsymbol{A}$: solve $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{x}=\boldsymbol{A}^{\mathrm{T}}\boldsymbol{w}$, then $\boldsymbol{w}_p=\boldsymbol{A}\boldsymbol{x}$
- orthonormal columns $\boldsymbol{Q}$: $\boldsymbol{w}_p=\boldsymbol{Q}\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{w}$
- hyperplane $\boldsymbol{n}\cdot\boldsymbol{x}=0$: $\boldsymbol{w}_p=\boldsymbol{w}-\frac{\boldsymbol{w}\cdot\boldsymbol{n}}{\boldsymbol{n}\cdot\boldsymbol{n}}\boldsymbol{n}$

$\boldsymbol{w}-\boldsymbol{w}_p$ is orthogonal to V (it is the projection onto $V^{\perp}$); $\|\boldsymbol{w}-\boldsymbol{w}_p\|$ is the distance from $\boldsymbol{w}$ to V; $\boldsymbol{w}\in V\Leftrightarrow\boldsymbol{w}_p=\boldsymbol{w}$.

Projection matrix $\boldsymbol{P}=\boldsymbol{A}(\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A})^{-1}\boldsymbol{A}^{\mathrm{T}}$: symmetric, $\boldsymbol{P}^2=\boldsymbol{P}$, eigenvalues 1 (on V) and 0 (on $V^{\perp}$). Line $\mathrm{span}\{\boldsymbol{u}\}$: $\boldsymbol{P}=\boldsymbol{u}\boldsymbol{u}^{\mathrm{T}}/\boldsymbol{u}^{\mathrm{T}}\boldsymbol{u}$.

Gram–Schmidt $\boldsymbol{v}_1=\boldsymbol{u}_1$, then
$$\boldsymbol{v}_j=\boldsymbol{u}_j-\sum_{i<j}\frac{\boldsymbol{u}_j\cdot\boldsymbol{v}_i}{\boldsymbol{v}_i\cdot\boldsymbol{v}_i}\boldsymbol{v}_i,\quad \boldsymbol{q}_j=\frac{\boldsymbol{v}_j}{\|\boldsymbol{v}_j\|}.$$
Subtract along the new $\boldsymbol{v}_i$, never the old $\boldsymbol{u}_i$. Rescale any $\boldsymbol{v}_j$ freely to clear fractions. $\boldsymbol{v}_j=\boldsymbol{0}\Leftrightarrow\boldsymbol{u}_j$ depends on the earlier vectors: drop it.

QR (independent columns) ❶ Gram–Schmidt on the columns of $\boldsymbol{A}$ gives $\boldsymbol{Q}=(\boldsymbol{q}_1\cdots\boldsymbol{q}_n)$. ❷ $\boldsymbol{R}=\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{A}$: upper triangular, $r_{ij}=\boldsymbol{q}_i\cdot\boldsymbol{a}_j$, $r_{ii}>0$. Check $\boldsymbol{Q}\boldsymbol{R}=\boldsymbol{A}$.

Orthogonal matrix (square) $\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{Q}=\boldsymbol{I}\Leftrightarrow\boldsymbol{Q}^{-1}=\boldsymbol{Q}^{\mathrm{T}}\Leftrightarrow$ columns orthonormal $\Leftrightarrow$ rows orthonormal. $\det=\pm1$; $\|\boldsymbol{Q}\boldsymbol{x}\|=\|\boldsymbol{x}\|$; $\boldsymbol{Q}\boldsymbol{x}\cdot\boldsymbol{Q}\boldsymbol{y}=\boldsymbol{x}\cdot\boldsymbol{y}$; products and inverses stay orthogonal. Non-square with $\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{Q}=\boldsymbol{I}$: $\boldsymbol{Q}\boldsymbol{Q}^{\mathrm{T}}\neq\boldsymbol{I}$.

5 Least squares

$\hat{\boldsymbol{x}}$ is a least squares solution of $\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}\Leftrightarrow\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}\hat{\boldsymbol{x}}=\boldsymbol{A}^{\mathrm{T}}\boldsymbol{b}\Leftrightarrow\boldsymbol{A}\hat{\boldsymbol{x}}$ is the projection of $\boldsymbol{b}$ onto $\mathrm{Col}(\boldsymbol{A})$.

Solving the normal equations are always consistent. **Unique** $\Leftrightarrow$ columns independent: $\hat{\boldsymbol{x}}=(\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A})^{-1}\boldsymbol{A}^{\mathrm{T}}\boldsymbol{b}$, or with QR backsubstitute $\boldsymbol{R}\hat{\boldsymbol{x}}=\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{b}$. **Not unique:** every solution is $\hat{\boldsymbol{x}}_0+\mathrm{Null}(\boldsymbol{A})$, and $\boldsymbol{A}\hat{\boldsymbol{x}}$ is the same for all. Error $\|\boldsymbol{b}-\boldsymbol{A}\hat{\boldsymbol{x}}\|$. If $\boldsymbol{A}\boldsymbol{x}=\boldsymbol{b}$ is consistent, least squares solutions = solutions.

Curve fitting $y=c_0+c_1x+\cdots+c_dx^d$ through N points: row i of $\boldsymbol{A}$ is $(1,x_i,\dots,x_i^d)$, $\boldsymbol{b}=(y_i)$. For a line the normal equations are
$$\begin{pmatrix}N&\sum x_i\\ \sum x_i&\sum x_i^2\end{pmatrix}\begin{pmatrix}c_0\\c_1\end{pmatrix}=\begin{pmatrix}\sum y_i\\ \sum x_iy_i\end{pmatrix}.$$
Distinct x_i: an exact fit is guaranteed once $d\ge N-1$; for $d<N-1$ the least squares fit is unique.

6 Eigenvalues and diagonalization

Eigenvalues roots of $\det(x\boldsymbol{I}-\boldsymbol{A})$. Triangular: the diagonal. 2×2: $x^2-\mathrm{tr}(\boldsymbol{A})x+\det\boldsymbol{A}$. 3×3: $x^3-\mathrm{tr}(\boldsymbol{A})x^2+(M_{11}+M_{22}+M_{33})x-\det\boldsymbol{A}$, where M_{ii} is the 2×2 minor without row and column i. Check: $\sum\lambda_i=\mathrm{tr}\,\boldsymbol{A}$, $\prod\lambda_i=\det\boldsymbol{A}$.

Shortcuts every row (or every column) sums to $s\Rightarrow\lambda=s$. Block triangular $\Rightarrow$ eigenvalues of the diagonal blocks. Rank 1, $\boldsymbol{A}=\boldsymbol{u}\boldsymbol{v}^{\mathrm{T}}$: $\lambda=\boldsymbol{v}\cdot\boldsymbol{u}$ and 0 ($n-1$ times). Before expanding $\det(x\boldsymbol{I}-\boldsymbol{A})$, add rows or columns to pull out a common factor; integer roots divide $\det\boldsymbol{A}$.

Eigenspace $E_\lambda=\mathrm{Null}(\lambda\boldsymbol{I}-\boldsymbol{A})$: reduce and read off a basis. Geometric mult. $=\dim E_\lambda=n-\mathrm{rank}(\lambda\boldsymbol{I}-\boldsymbol{A})$; $1\le$ geometric $\le$ algebraic.

Eigen-data for free $\boldsymbol{A}^k$: λ^k ▪ $\boldsymbol{A}^{-1}$: $1/\lambda$ ▪ $\boldsymbol{A}+c\boldsymbol{I}$: $\lambda+c$ ▪ $p(\boldsymbol{A})$: $p(\lambda)$, all with the same eigenvectors. $\boldsymbol{A}^{\mathrm{T}}$: same eigenvalues. $\boldsymbol{A}^k=\boldsymbol{0}\Rightarrow$ only $\lambda=0$. $\boldsymbol{A}^2=\boldsymbol{A}\Rightarrow\lambda\in\{0,1\}$. $\boldsymbol{B}=\boldsymbol{P}^{-1}\boldsymbol{A}\boldsymbol{P}$: same characteristic polynomial, and $\boldsymbol{B}\boldsymbol{w}=\lambda\boldsymbol{w}\Rightarrow\boldsymbol{A}(\boldsymbol{P}\boldsymbol{w})=\lambda(\boldsymbol{P}\boldsymbol{w})$. Nonzero combinations inside one E_λ are eigenvectors; **across different λ they are not**.

Diagonalize ❶ Eigenvalues. ❷ A basis of every E_λ. ❸ Diagonalizable $\Leftrightarrow\sum\dim E_\lambda=n\Leftrightarrow$ the characteristic polynomial splits and geometric = algebraic for every λ. Only repeated λ need checking; n distinct eigenvalues $\Rightarrow$ yes. ❹ $\boldsymbol{P}=(\boldsymbol{v}_1\cdots\boldsymbol{v}_n)$, $\boldsymbol{D}=\mathrm{diag}(\lambda_1,\dots,\lambda_n)$ **in the same order**: $\boldsymbol{A}=\boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{-1}$.

With parameters for each repeated λ demand $\mathrm{rank}(\lambda\boldsymbol{I}-\boldsymbol{A})=n-$ (algebraic mult.); treat separately the parameter values that make two eigenvalues coincide. Never diagonalizable: $\begin{pmatrix}1&1\\0&1\end{pmatrix}$, $\begin{pmatrix}0&1\\0&0\end{pmatrix}$. Only one eigenvalue λ: diagonalizable $\Leftrightarrow\boldsymbol{A}=\lambda\boldsymbol{I}$.

Powers $\boldsymbol{A}^m=\boldsymbol{P}\boldsymbol{D}^m\boldsymbol{P}^{-1}$; a square root is $\boldsymbol{P}\boldsymbol{D}^{1/2}\boldsymbol{P}^{-1}$. For $\boldsymbol{A}^m\boldsymbol{x}_0$ write $\boldsymbol{x}_0=\sum c_i\boldsymbol{v}_i$ (solve $\boldsymbol{P}\boldsymbol{c}=\boldsymbol{x}_0$): $\boldsymbol{A}^m\boldsymbol{x}_0=\sum c_i\lambda_i^m\boldsymbol{v}_i$. As $m\to\infty$ the $|\lambda|<1$ terms vanish and $\lambda=1$ stays. Cayley–Hamilton: $p(\boldsymbol{A})=\boldsymbol{0}$ for $p(x)=\det(x\boldsymbol{I}-\boldsymbol{A})$.

Orthogonal diagonalization exists $\Leftrightarrow\boldsymbol{A}$ symmetric. ❶ Eigenspaces. ❷ Gram–Schmidt **inside** each E_λ of dimension ≥2 (different eigenspaces are already orthogonal). ❸ Normalize: $\boldsymbol{P}$ orthogonal, $\boldsymbol{A}=\boldsymbol{P}\boldsymbol{D}\boldsymbol{P}^{\mathrm{T}}=\sum\lambda_i\boldsymbol{q}_i\boldsymbol{q}_i^{\mathrm{T}}$. Symmetric $\Rightarrow$ diagonalizable; **diagonalizable $\not\Rightarrow$ symmetric**.

6 Markov chains

Stochastic $\boldsymbol{P}$ entries ≥0, **columns sum to 1**; p_{ij} is the probability of moving **from** j **to** i. $\boldsymbol{x}_k=\boldsymbol{P}^k\boldsymbol{x}_0$; k-step probabilities are the entries of $\boldsymbol{P}^k$. Eigenvalues: 1 always, all $|\lambda|\le1$. Web surfer: column j has $1/($number of links out of page $j)$ in the rows of the pages it links to.

Equilibrium solve $(\boldsymbol{P}-\boldsymbol{I})\boldsymbol{x}=\boldsymbol{0}$ and scale so the entries sum to 1. Regular (some $\boldsymbol{P}^k$ has all entries >0) $\Rightarrow$ the equilibrium is unique and every chain converges to it.
EG $\begin{pmatrix}1-a&b\\a&1-b\end{pmatrix}$: equilibrium $1/(a+b)\begin{pmatrix}b\\a\end{pmatrix}$, other eigenvalue $1-a-b$.

6 Singular value decomposition

SVD $\boldsymbol{A}=\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^{\mathrm{T}}$ with $\boldsymbol{U}$ ($m\times m$) and $\boldsymbol{V}$ ($n\times n$) orthogonal, $\boldsymbol{\Sigma}$ $m\times n$.

❶ $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}$ (entries $\boldsymbol{a}_i\cdot\boldsymbol{a}_j$): eigenvalues $\mu_1\ge\cdots\ge\mu_n\ge0$; $\sigma_i=\sqrt{\mu_i}$; r of them are nonzero.

❷ $\boldsymbol{V}=(\boldsymbol{v}_1\cdots\boldsymbol{v}_n)$: orthonormal eigenvectors of $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}$ **in that order** (Gram–Schmidt inside repeated eigenspaces).

❸ $\boldsymbol{u}_i=\boldsymbol{A}\boldsymbol{v}_i/\sigma_i$ for $i\le r$.

❹ If $r<m$: complete $\boldsymbol{U}$ with an orthonormal basis of $\mathrm{Null}(\boldsymbol{A}^{\mathrm{T}})$.

❺ $\boldsymbol{\Sigma}$ has $\sigma_1,\dots,\sigma_r$ down the diagonal, zeros elsewhere.

Wide matrix ($m<n$) factor $\boldsymbol{A}^{\mathrm{T}}=\boldsymbol{U}'\boldsymbol{\Sigma}'^{\mathrm{T}}\boldsymbol{V}'^{\mathrm{T}}$, then $\boldsymbol{A}=\boldsymbol{V}'\boldsymbol{\Sigma}'^{\mathrm{T}}\boldsymbol{U}'^{\mathrm{T}}$. Check any SVD with $\boldsymbol{A}\boldsymbol{v}_i=\sigma_i\boldsymbol{u}_i$.

Reading an SVD rank $=r$; $\boldsymbol{A}=\sum_{i\le r}\sigma_i\boldsymbol{u}_i\boldsymbol{v}_i^{\mathrm{T}}$. Orthonormal bases: $\boldsymbol{u}_1,\dots,\boldsymbol{u}_r$ of $\mathrm{Col}(\boldsymbol{A})$; $\boldsymbol{u}_{r+1},\dots,\boldsymbol{u}_m$ of $\mathrm{Null}(\boldsymbol{A}^{\mathrm{T}})$; $\boldsymbol{v}_1,\dots,\boldsymbol{v}_r$ of $\mathrm{Row}(\boldsymbol{A})$; $\boldsymbol{v}_{r+1},\dots,\boldsymbol{v}_n$ of $\mathrm{Null}(\boldsymbol{A})$. Projection onto $\mathrm{Col}(\boldsymbol{A})$: $\sum_{i\le r}(\boldsymbol{w}\cdot\boldsymbol{u}_i)\boldsymbol{u}_i$. Eigenvalues of $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}$: the σ_i^2 and $n-r$ zeros; of $\boldsymbol{A}\boldsymbol{A}^{\mathrm{T}}$: the σ_i^2 and $m-r$ zeros.

7 Linear transformations $T:\mathbb{R}^n\to\mathbb{R}^m$

Linear? show $T(\boldsymbol{x})=\boldsymbol{A}\boldsymbol{x}$ by writing $\boldsymbol{A}$, or check $T(a\boldsymbol{u}+b\boldsymbol{v})=aT(\boldsymbol{u})+bT(\boldsymbol{v})$. **Not linear:** $T(\boldsymbol{0})\neq\boldsymbol{0}$, added constants, products, powers or $|\cdot|$ of coordinates.

Standard matrix $\boldsymbol{A}=(T(\boldsymbol{e}_1)\cdots T(\boldsymbol{e}_n))$. $S\circ T$ has matrix $\boldsymbol{B}\boldsymbol{A}$ ($\boldsymbol{B}$ that of S).

From the images of a basis $S=\{\boldsymbol{u}_i\}$ $[T]_S=(T(\boldsymbol{u}_1)\cdots T(\boldsymbol{u}_n))$, $T(\boldsymbol{w})=[T]_S[\boldsymbol{w}]_S$, $\boldsymbol{A}=[T]_S\boldsymbol{U}^{-1}$ with $\boldsymbol{U}=(\boldsymbol{u}_1\cdots\boldsymbol{u}_n)$. By hand: $(\boldsymbol{U}^{\mathrm{T}}\mid[T]_S^{\mathrm{T}})\to(\boldsymbol{I}\mid\boldsymbol{A}^{\mathrm{T}})$.

Images on a spanning set determine T; on a merely independent set **they do not**.

Range and kernel $R(T)=\mathrm{Col}(\boldsymbol{A})=\mathrm{Col}([T]_S)$; $\ker T=\mathrm{Null}(\boldsymbol{A})\neq\mathrm{Null}([T]_S)$. $\mathrm{rank}\,T+\mathrm{nullity}\,T=n$.

Injective $\Leftrightarrow\ker T=\{\boldsymbol{0}\}\Leftrightarrow$ rank n. **Surjective** $\Leftrightarrow R(T)=\mathbb{R}^m\Leftrightarrow$ rank m. $n=m$: injective $\Leftrightarrow$ surjective $\Leftrightarrow\det\boldsymbol{A}\neq0$.

Geometry rotation by θ: $\begin{pmatrix}\cos\theta&-\sin\theta\\ \sin\theta&\cos\theta\end{pmatrix}$; with the reflections $\begin{pmatrix}\cos\theta&\sin\theta\\ \sin\theta&-\cos\theta\end{pmatrix}$ these are all 2×2 orthogonal matrices. Projection onto V: $\boldsymbol{P}$ (symmetric, $\boldsymbol{P}^2=\boldsymbol{P}$); reflection in V: $2\boldsymbol{P}-\boldsymbol{I}$.

Fixed vectors and limits $T(\boldsymbol{u})=\boldsymbol{u}$: solve $(\boldsymbol{A}-\boldsymbol{I})\boldsymbol{u}=\boldsymbol{0}$. $T(\boldsymbol{u})=\boldsymbol{b}$: solve $\boldsymbol{A}\boldsymbol{u}=\boldsymbol{b}$. A basis with $T(\sum c_i\boldsymbol{u}_i)=\sum c_i\lambda_i\boldsymbol{u}_i$ exists $\Leftrightarrow\boldsymbol{A}$ is diagonalizable with eigenpairs $(\lambda_i,\boldsymbol{u}_i)$. $\lim T^k(\boldsymbol{x})$: expand $\boldsymbol{x}$ in eigenvectors, keep the $\lambda=1$ part.

Notation $\boldsymbol{A}$ is $m\times n$ with columns $\boldsymbol{a}_j$ · $r=\mathrm{rank}\,\boldsymbol{A}$ · $\boldsymbol{R}$ is a row-echelon form of $\boldsymbol{A}$ · vectors are columns · chips give the lecture chapter